

Week 8 — Lecture notes

The Laplace transform

The figures below are snapshots of interactive widgets. The live versions, where you can move the sliders yourself, are at tchaffey.com/elec2302.

Lecture outline

1. Motivation: unstable systems
2. Making the Fourier transform converge — the bilateral Laplace transform.
3. Least-squares approximation and inverse Laplace transform.
4. Initial conditions and the unilateral Laplace transform.
5. Example
6. Laplace transform properties.

Motivation: unstable systems

Last week we saw that exponential and marginal stability can be determined from the system poles:

- poles in LHP: stable
- poles (non-repeated) on imaginary axis: marginally stable.

For a single left half plane pole, we have

$$\frac{1}{p + j\omega} \xrightarrow[\text{F.T.}]{\text{Inverse}} e^{-pt} H(t)$$

We would guess that, for a right half plane pole,

$$\frac{1}{-p + j\omega} \rightsquigarrow e^{pt} H(t).$$

But: The Fourier transform of $e^{pt} H(t)$ is

$$\int_0^\infty e^{pt} e^{-j\omega t} dt = \infty \dots$$

We need a trick to make this converge, and that is the Laplace transform.

Remember what the Fourier transform is: for each ω ,

$$\int_{-\infty}^\infty u(t) e^{-j\omega t} dt$$

projects $u(t)$ onto $e^{+j\omega t}$.

If $u(t)$ decays, this converges. If $u(t)$ grows, it does not.

Making the Fourier transform converge: the bilateral Laplace transform

The trick in the Laplace transform is to change the basis function: choose a basis function that decays at least as fast as $u(t)$ grows, then the projection converges.

For $u(t) = e^{pt}H(t)$, what can we do?

Shift $e^{j\omega t}$ to the right:

$$e^{(\sigma+j\omega)t} = e^{\sigma t}e^{j\omega t}.$$

Then the projection becomes

$$\int_{-\infty}^{\infty} e^{pt}e^{-\sigma t}e^{-j\omega t}H(t) dt = \int_0^{\infty} e^{(p-\sigma)t}e^{-j\omega t} dt.$$

If $\sigma > p$, this converges:

$$\begin{aligned} \int_0^{\infty} e^{(p-\sigma)t}e^{-j\omega t} dt &= \left[\frac{-1}{\sigma - p + j\omega} e^{(p-\sigma-j\omega)t} \right]_0^{\infty} \\ &= \frac{1}{-p + \sigma + j\omega} \\ &= \frac{1}{-p + s}, \quad \text{where } s = \sigma + j\omega. \end{aligned}$$

This shifted Fourier transform is called the bilateral Laplace transform:

$$\mathcal{B}(u(t))(s) = \int_{-\infty}^{\infty} u(t)e^{-st} dt$$

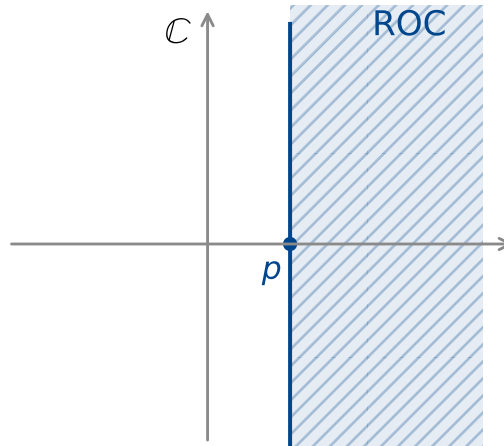


Figure 1: Region of convergence for $u(t) = e^{pt}H(t)$.

$\sigma > p$ defines the Region of Convergence (ROC), the set $s \in \mathbb{C}$ such that, for a given $u(t)$,

$$\int_{-\infty}^{\infty} u(t)e^{-st} dt$$

converges.

For a fixed $\sigma \in \text{ROC}$, $\mathcal{B}$ is the Fourier transform of the weighted signal $u(t)e^{-\sigma t}$. Fixing ω , $\mathcal{B}$ projects this weighted signal onto the basis function $e^{+j\omega t}$.

If the region of convergence contains the imaginary axis (the signal decays), then we can take $\sigma = 0$ and

$$\mathcal{B}(u) = \int_{-\infty}^{\infty} u(t)e^{-j\omega t} dt$$

which is precisely the Fourier transform.

Least-squares approximation and the inverse Laplace transform

Given a signal $u(t)$ and its Laplace transform $U(s)$, as well as its region of convergence, we can find the corresponding time domain signal by fixing $\sigma \in \text{ROC}$ and computing the inverse Fourier transform of the weighted signal $u(t)e^{-\sigma t}$.

Note first that the Fourier transform of $u(t)e^{-\sigma t}$ is $U(j\omega + \sigma)$ from the Fourier transform properties.

Now, taking the inverse Fourier transform,

$$u(t)e^{-\sigma t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega + \sigma) e^{j\omega t} d\omega.$$

Divide by $e^{-\sigma t}$ to get

$$u(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega + \sigma) e^{(\sigma + j\omega)t} d\omega$$

This is the Inverse Laplace Transform. Any $\sigma \in \text{ROC}$ can be used.

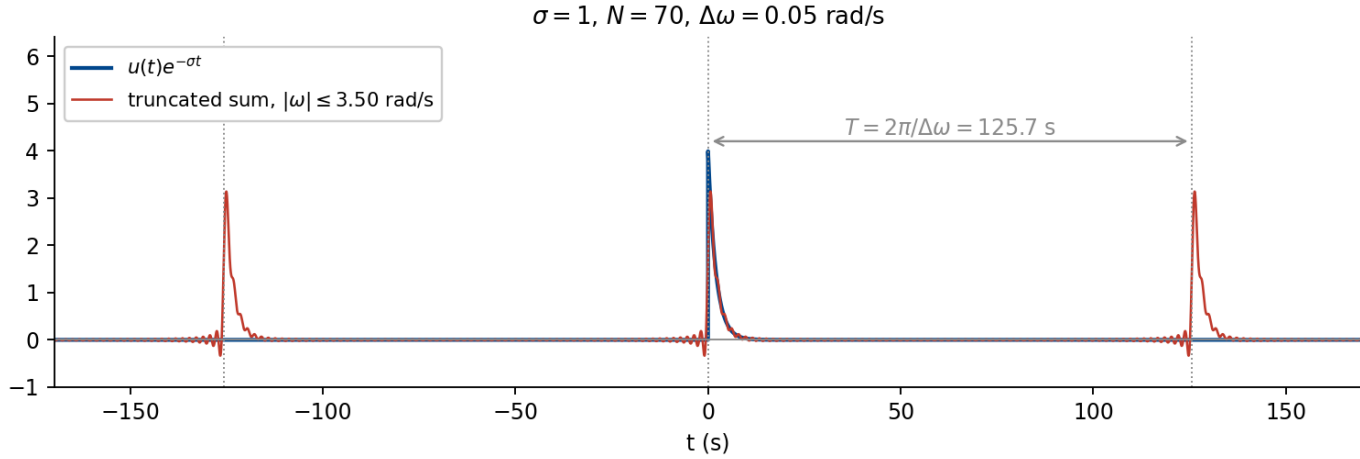


Figure 3: The weighted signal $u(t)e^{-\sigma t}$ and its truncated Fourier sum over a wider time window. Discretising the inversion integral on $\omega_k = k\Delta\omega$ makes the reconstruction periodic with period $T = 2\pi/\Delta\omega$.

Initial conditions and the unilateral Laplace transform

Bilateral refers to the fact that we integrate from $t = -\infty$ to $t = \infty$. Often, however, we're only interested in a system after we turn it on at $t = 0$, we don't want to have to keep track of the infinite past. Also, the impulse response of a real system starts at $t = 0$ (that is, the system is causal, and the output depends only on the past, not the future). This motivates the unilateral Laplace transform, which only looks at what happens from $t = 0^- = 0 - \varepsilon$.

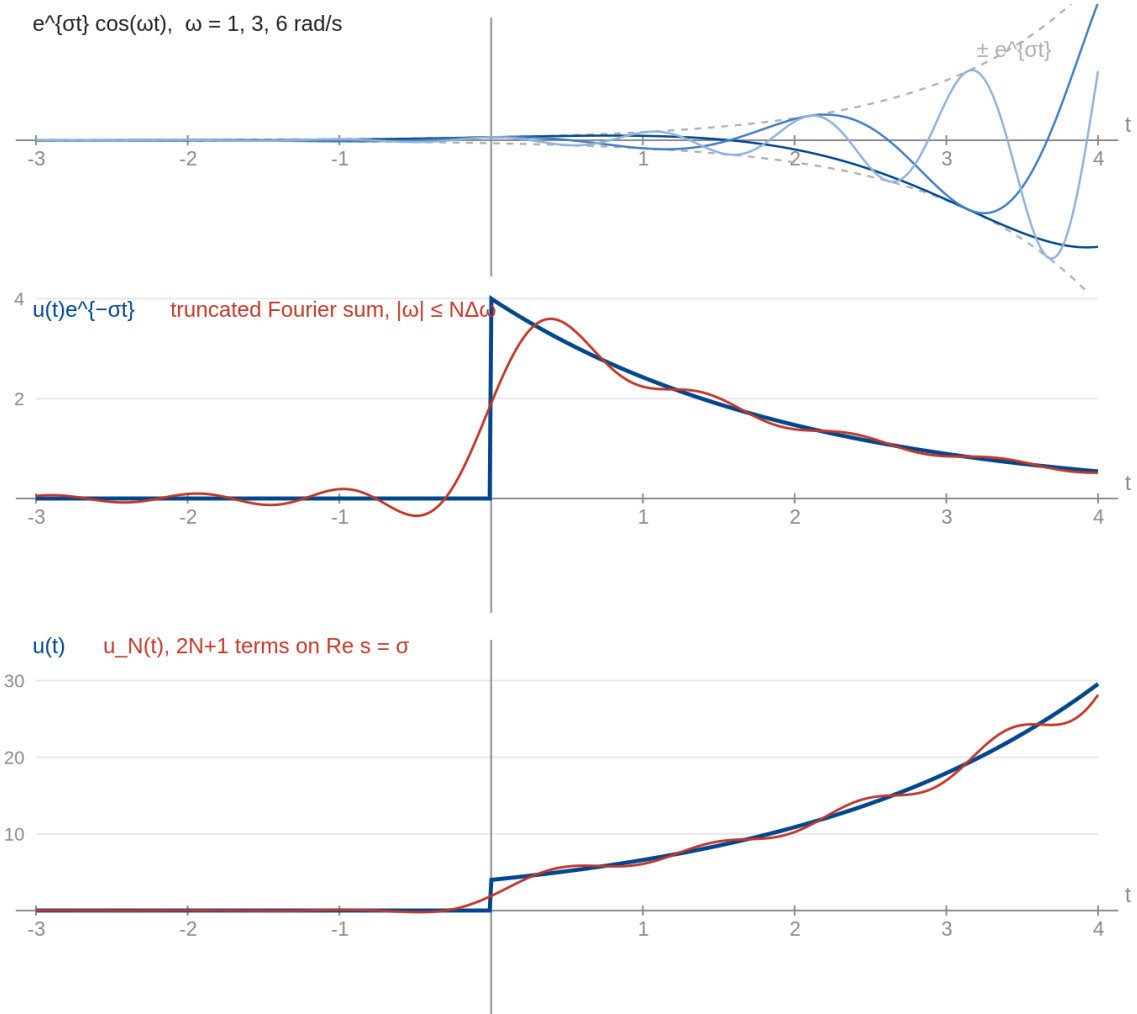


Figure 2: The expansion on $\text{Re } s = \sigma$ at $\sigma = 1.00$ and $N = 130$, so 261 terms spaced $\Delta\omega = 0.05$ rad/s apart over $|\omega| \leq 6.50$ rad/s. Top: three of the basis signals $e^{\sigma t} \cos \omega t$ at $\omega = 1, 3, 6$ rad/s, inside the envelope $\pm e^{\sigma t}$. Middle: the weighted signal $u(t)e^{-\sigma t}$ and the truncated sum. Bottom: $u(t)$ and the reconstruction $u_N(t)$.

$$\mathcal{L}(u(t)) = \int_{0^-}^{\infty} u(t) e^{-st} dt = \mathcal{B}(u(t)H(t)),$$

The second equality requires u continuous at 0.

This transform lets us account for initial conditions as well as inputs. Let's see how by looking at the Laplace transform of the derivative.

$$\int_0^{\infty} \frac{d}{dt} u(t) e^{-st} dt.$$

Integrate by parts:

$$= u(t)e^{-st} \Big|_0^{\infty} - \int_0^{\infty} -s u(t) e^{-st} dt.$$

$\lim_{t \rightarrow \infty} e^{-st} = 0$ for $s \in \text{ROC}$, so we have

$$= -u(0) + s U(s).$$

What about the second derivative?

Let $v(t) = \frac{d}{dt} u(t)$.

Then

$$\begin{aligned} \mathcal{L}(v(t)) &= -v(0) + s V(s) \\ &= -\frac{d}{dt} u(0) - s u(0) + s^2 U(s). \end{aligned}$$

By induction,

$$\mathcal{L}(u^{(n)}(t)) = -u^{(n-1)}(0) - s u^{(n-2)}(0) - \dots - s^{n-1} u(0) + s^n U(s)$$

As with the Fourier transform, we have a table of standard Laplace transforms. The derivations are very similar to those for the Fourier transforms.

$u(t)$	$U(s)$	ROC
$\delta(t)$	1	all s
$H(t)$	$\frac{1}{s}$	$\text{Re}(s) > 0$
$t H(t)$	$\frac{1}{s^2}$	$\text{Re}(s) > 0$
$e^{-at} H(t)$	$\frac{1}{s+a}$	$\text{Re}(s) > -a$
$t e^{-at} H(t)$	$\frac{1}{(s+a)^2}$	$\text{Re}(s) > -a$
$\cos(\omega_0 t) H(t)$	$\frac{s}{s^2 + \omega_0^2}$	$\text{Re}(s) > 0$
$\sin(\omega_0 t) H(t)$	$\frac{\omega_0}{s^2 + \omega_0^2}$	$\text{Re}(s) > 0$
$e^{-at} \cos(\omega_0 t) H(t)$	$\frac{s+a}{(s+a)^2 + \omega_0^2}$	$\text{Re}(s) > -a$
$e^{-at} \sin(\omega_0 t) H(t)$	$\frac{\omega_0}{(s+a)^2 + \omega_0^2}$	$\text{Re}(s) > -a$

Example

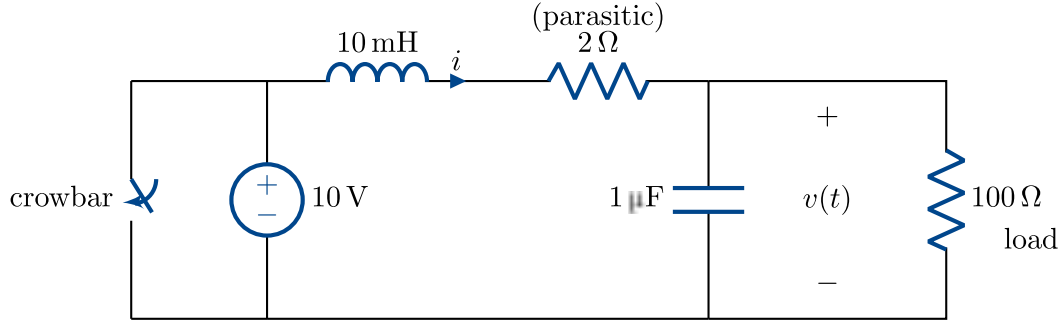


Figure 4: Supply feeding a resistive load through an LC filter with parasitic resistance, with a crowbar overload protector across the source.

A crowbar is an overvoltage protector, which shorts the source if the load voltage exceeds a given threshold. Suppose the circuit is at steady state and the crowbar is activated at $t = 0$. What is the capacitor voltage $v(t)$ for $t > 0$?

The current splits through the capacitor and load:

$$i = C\dot{v} + \frac{v}{R_L}.$$

It is equal to the current through the inductor and resistor.

$$i = \frac{1}{R}v_R.$$

$$v_L = L\frac{d}{dt}i$$

KVL around the loop:

$$v(t) + v_L + v_R = 0.$$

Combining:

$$v + LC\ddot{v} + \frac{L}{R_L}\dot{v} + RC\dot{v} + \frac{R}{R_L}v = 0 \quad \text{source is shorted}$$

$$LC\ddot{v} + \left(\frac{L}{R_L} + RC\right)\dot{v} + \left(1 + \frac{R}{R_L}\right)v = 0.$$

Take the Laplace transform:

$$\begin{aligned} LC(-\dot{v}(0) - s v(0) + s^2 V(s)) + \left(\frac{L}{R_L} + RC\right)(-v(0) + s V(s)) \\ + \left(1 + \frac{R}{R_L}\right)V(s) = 0. \end{aligned}$$

Circuit is at steady state at $t = 0$. At DC the inductor is a short and the capacitor carries no current, so the source drives R and R_L in series:

$$v(0) = 10 \frac{R_L}{R + R_L} = 10 \times \frac{100}{102} = 9.804 \text{ V}, \quad i(0) = \frac{10}{R + R_L} = 98.04 \text{ mA}.$$

The capacitor current is $i(0) - v(0)/R_L = 0$, so $\dot{v}(0) = 0$.

$$s^2 LC V(s) + \left(\frac{L}{R_L} + RC \right) s V(s) + \left(1 + \frac{R}{R_L} \right) V(s) = LC s v(0) + \left(\frac{L}{R_L} + RC \right) v(0)$$

$$V(s) = \frac{LC s v(0) + \left(\frac{L}{R_L} + RC \right) v(0)}{s^2 LC + \left(\frac{L}{R_L} + RC \right) s + \left(1 + \frac{R}{R_L} \right)}$$

We now want to inverse transform this expression by breaking it into parts that match entries in the Laplace transform table.

Normalise so the coefficient of s^2 is 1:

$$V(s) = \frac{v(0) s + v(0) \left(\frac{1}{CR_L} + \frac{R}{L} \right)}{s^2 + \left(\frac{1}{CR_L} + \frac{R}{L} \right) s + \left(\frac{1}{LC} + \frac{R}{R_L LC} \right)}$$

Denote

$$2\alpha = \frac{1}{R_L C} + \frac{R}{L}, \quad \omega_0^2 = \frac{1 + R/R_L}{LC}.$$

With the values above, $\alpha = 5100 \text{ s}^{-1}$ and $\omega_0 \approx 10100$.

We now have

$$V(s) = \frac{v(0) s + 2\alpha v(0)}{s^2 + 2\alpha s + \omega_0^2}$$

Is the discriminant > 0 or < 0 ?

$$4\alpha^2 - 4\omega_0^2 = 4(\alpha - \omega_0)(\alpha + \omega_0).$$

$\alpha < \omega_0$, so the system is underdamped.

Complete the square in the denominator:

$$s^2 + 2\alpha s + \omega_0^2 = (s + \alpha)^2 + (\omega_0^2 - \alpha^2).$$

Let $\omega^2 = \omega_0^2 - \alpha^2$.

Then

$$\begin{aligned} V(s) &= \frac{v(0)(s + \alpha) + v(0) \alpha}{(s + \alpha)^2 + \omega^2} \\ &= \frac{v(0)(s + \alpha)}{(s + \alpha)^2 + \omega^2} + \frac{v(0) \alpha}{\omega} \frac{\omega}{(s + \alpha)^2 + \omega^2}. \end{aligned}$$

Now use the table entries

$$e^{-\alpha t} \cos \omega t \rightsquigarrow \frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$$

and

$$e^{-\alpha t} \sin \omega t \rightsquigarrow \frac{\omega}{(s + \alpha)^2 + \omega^2}$$

to get

$$v(t) = e^{-\alpha t} \left(v(0) \cos \omega t + \frac{v(0) \alpha}{\omega} \sin \omega t \right) H(t).$$

With numbers,

$$v(t) = e^{-5100t} (9.804 \cos 8717t + 5.736 \sin 8717t) H(t).$$

Properties of the Laplace transform

The Laplace transform inherits many useful properties of the Fourier transform. We have already seen the derivative property. Let's now check convolution.

Let $y(t) = h(t) * u(t)$. Suppose $h(t) = u(t) = 0$ for $t < 0$.

$$\begin{aligned} h(t) * u(t) &= \int_{-\infty}^{\infty} h(t - \tau) u(\tau) d\tau \\ &= \int_0^t h(t - \tau) u(\tau) d\tau. \end{aligned}$$

$$\mathcal{L}(h(t) * u(t)) = \int_0^{\infty} \int_0^t h(t - \tau) u(\tau) d\tau e^{-st} dt.$$

If we are in the ROC of both signals, we can swap the order of integration.

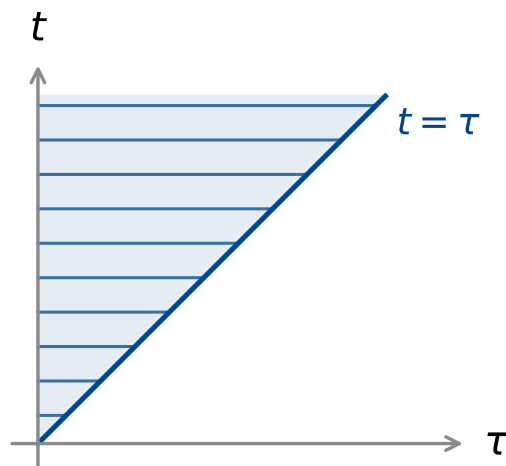


Figure 5: t from 0 to ∞ , τ from 0 to t .

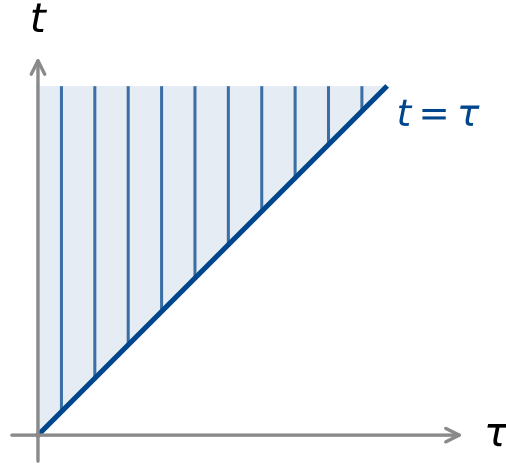


Figure 6: τ from 0 to ∞ , t from τ to ∞ .

$$= \int_0^\infty \int_\tau^\infty h(t-\tau) e^{-st} dt u(\tau) d\tau \quad (\text{allowed in the ROC}).$$

Let $\mu = t - \tau$.

$$d\mu = dt$$

$$t = \infty \Rightarrow \mu = \infty$$

$$t = \tau \Rightarrow \mu = 0.$$

$$\begin{aligned} &= \int_0^\infty \int_0^\infty h(\mu) e^{-s(\mu+\tau)} d\mu u(\tau) d\tau \\ &= \int_0^\infty h(\mu) e^{-s\mu} d\mu \int_0^\infty u(\tau) e^{-s\tau} d\tau \\ &= H(s) U(s). \end{aligned}$$

Integration

We can use the convolution property to get the Laplace transform of an integrator:

$$\begin{aligned} y(t) &= \int_0^t u(\tau) d\tau \\ &= \int_0^\infty H(t-\tau) u(\tau) d\tau \\ &= H * u. \end{aligned}$$

So $Y(s) = H(s)U(s)$.

What is the Laplace transform of $H(t)$?

$$\begin{aligned}\int_0^\infty H(t)e^{-st} dt &= \int_0^\infty e^{-st} dt \\ &= \frac{-1}{s} e^{-st} \Big|_0^\infty = \frac{1}{s}.\end{aligned}$$

So

$$\int_0^t u(\tau) d\tau \rightsquigarrow \frac{1}{s} U(s).$$

Other properties

	$u(t)$	$U(s)$
linearity	$\alpha u_1(t) + \beta u_2(t)$	$\alpha U_1(s) + \beta U_2(s)$
time shift	$u(t-\tau)H(t-\tau)$	$e^{-s\tau}U(s)$
frequency shift	$e^{at}u(t)$	$U(s-a)$

Initial value theorem

$$\mathcal{L}\left(\frac{d}{dt}u(t)\right)(s) = sU(s) - u(0).$$

So

$$sU(s) = u(0) + \int_0^\infty u'(t)e^{-st} dt.$$

Now, if $|u'(t)| \leq Me^{ct}$ for some $M, c > 0$, the rightmost term $\rightarrow 0$ as $s \rightarrow \infty$ along the real axis, so we have

$$\lim_{s \rightarrow \infty} sU(s) = u(0).$$

Final value theorem

Again take

$$sU(s) = u(0) + \int_0^\infty u'(t)e^{-st} dt.$$

Let $s \rightarrow 0$ along the real axis. Then

$$\int_0^\infty u'(t)e^{-st} dt = \int_0^\infty u'(t) dt = \lim_{T \rightarrow \infty} u(T) - u(0).$$

So

$$\begin{aligned}\lim_{s \rightarrow 0} sU(s) &= \lim_{T \rightarrow \infty} u(T) - u(0) + u(0) \\ &= \lim_{T \rightarrow \infty} u(T), \quad \underline{\text{provided this limit exists.}}\end{aligned}$$